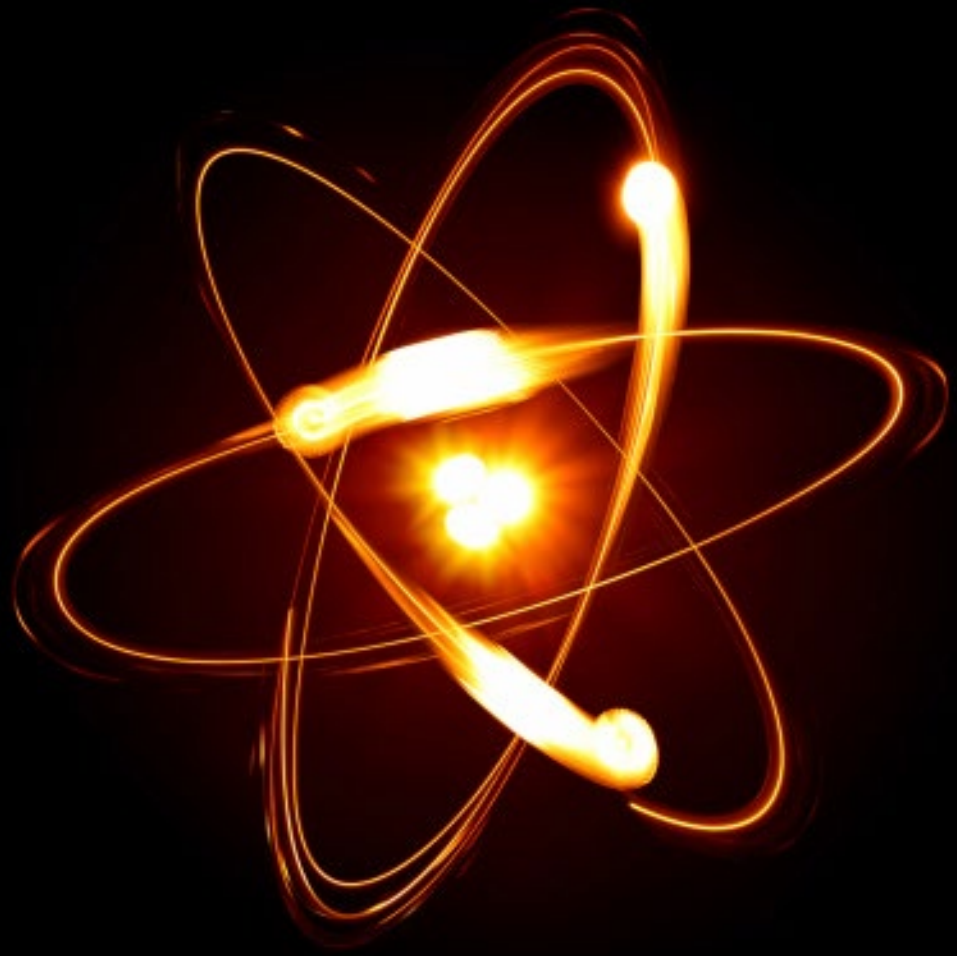
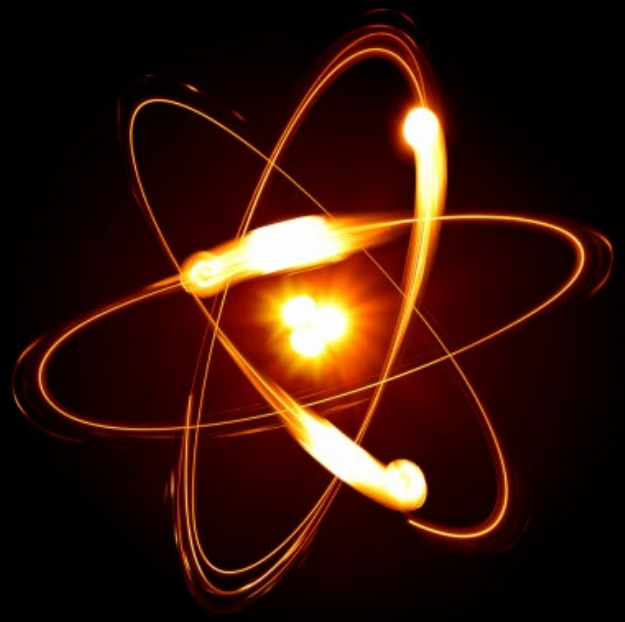
Chapter 4.

Models of the Atom



Beatriz González del Río

Quantum Physics



Chapter 4.

Models of the Atom

- Historic development (until XX century)
- XX century: new experimental evidence
- Nagaoka´s atomic model
- Thomson´s atomic model
- Rutherford experiment: scattering of α particles
- Rutherford´s atomic model
- Rutherford´s formula: size of nucleus, stability & anomalies
- Atomic spectra

Atomic Models

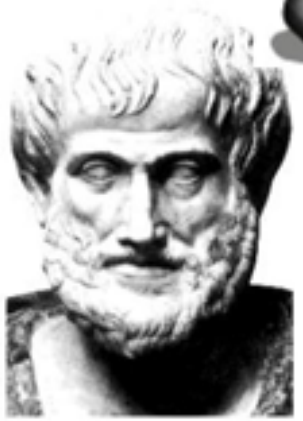
Early Greek Theories



Democritus

- 400 B.C. - Democritus thought matter could not be divided indefinitely.
- This led to the idea of atoms in a void.





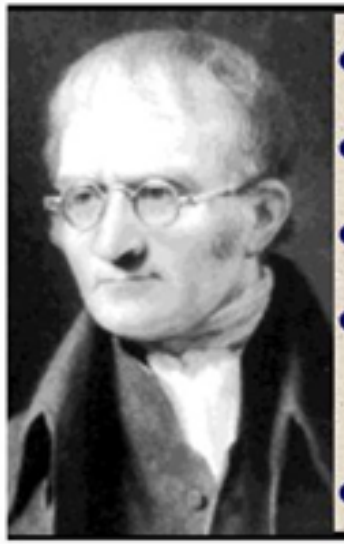
Aristotle

- 350 B.C - Aristotle modified an earlier theory that matter was made of four “elements”: earth, fire, water, air.
- Aristotle was wrong. However, his theory persisted for 2000 years.

Atomic Models

John Dalton

- 1800 -Dalton proposed a modern atomic model based on experimentation not on pure reason.



- All matter is made of atoms.
- Atoms of an element are identical.
- Each element has different atoms.
- Atoms of different elements combine in constant ratios to form compounds.
- Atoms are rearranged in reactions.

- His ideas account for the law of conservation of mass (atoms are neither created nor destroyed) and the law of constant composition (elements combine in fixed ratios).

Atomic Models

At the beginning of the XX century, the experimental evidence suggested that atoms contain electrons and the number of electrons was ≈ A/2 (where A is the atomic weight)

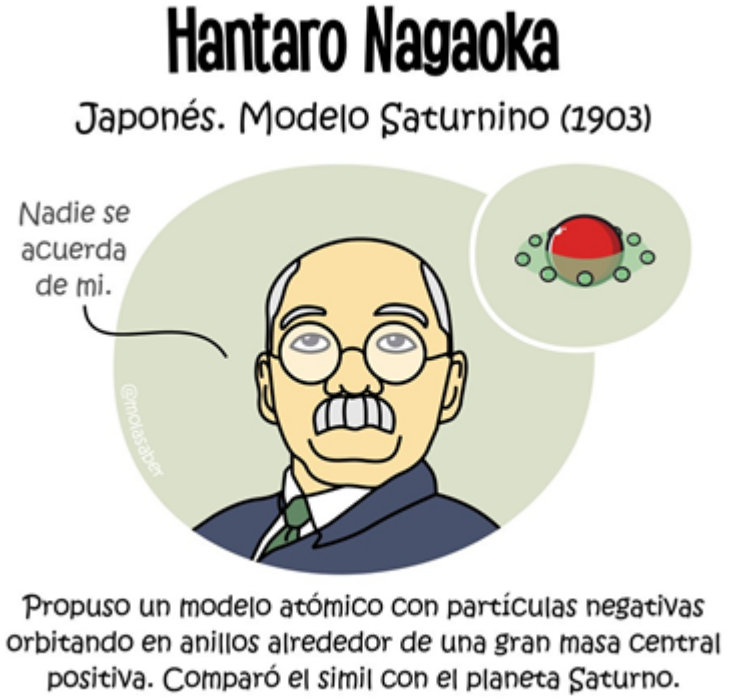
However, the atom is neutral → there must be some positive charge to neutralize the negative charge.

Moreover, m_e is very small → the atomic mass should be linked to the positive charge

How to arrange the positive and negative charges in the atom?

In 1903-04 two models are proposed: Nagaoka and Thomson

Nagaoka. Positive charge at the center and the electrons are located around (like Saturn's rings). However, the electrons must revolve around the positive charge and according to classical electromagnetic theory they must radiate energy so they would finally fall on the positive charge.



Thomson's Atomic Model

If the atom is neutral and contains very light negative electrons, how could the positive charge be arranged?

Thomson

Positive charge uniformly spread within a sphere with the atom's size. The electrons would be at rest uniformly distributed through the sphere of positive charge.

When an atom is excited, the electrons would vibrate about their equilibrium positions and they would emit electromagnetic radiation.

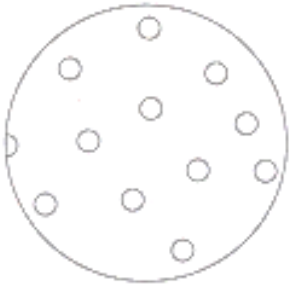


Figura 4.1. Esquema ilustrativo del átomo de Thomson. Los pequeños círculos representan los electrones, repartidos en una esfera con una distribución uniforme de carga positiva.

Thomson's Atomic Model. A calculation

Assume that there is one electron of charge -e at the center of a spherical region of uniform positive charge density ρ. Suppose that the electron is displaced to a distance r_0 from the center.

Show that its motion is a simple harmonic oscillation.

Gauss' flux theorem: The total electric flux through any closed surface is equal to $1/\epsilon_0$ times the net electric charge inside.

There is a force acting on the electron

$$F = -\frac{1}{4\pi\varepsilon_0}\left(\frac{4}{3}\pi\rho r_0^3\right)\frac{e}{r_0^2} = \frac{-\rho e r_0}{3\varepsilon_0} = -kr_0$$

Simple harmonic motion

and it will vibrate with a frequency

$$\nu = \frac{1}{2\pi}\sqrt{\frac{k}{m_e}}$$

For a H atom, we consider a total positive charge of one electron inside a sphere of radius $r_0 = 1.0$ Å. Then, we obtain $k = 2.3 \cdot 10^2$ N/m and we have $\nu \sim 2.5\ 10^{15}\ s^{-1}$ and $\lambda \sim 1200$ Å. This result is unrealistic !!

Scattering of α particles

Additional experimental evidence about the inadequacy of Thomson's model was provided by Rutherford as a result of some experiments with α particles

The α particles are doubly ionized helium atoms emitted spontaneously from several radioactive materials (Ra, U....)

Rutherford carried out a very extensive research program to study the properties of the α particles and the way they interact with matter.

Most interesting: Scattering of α particles on passing through thin foils of various materials.

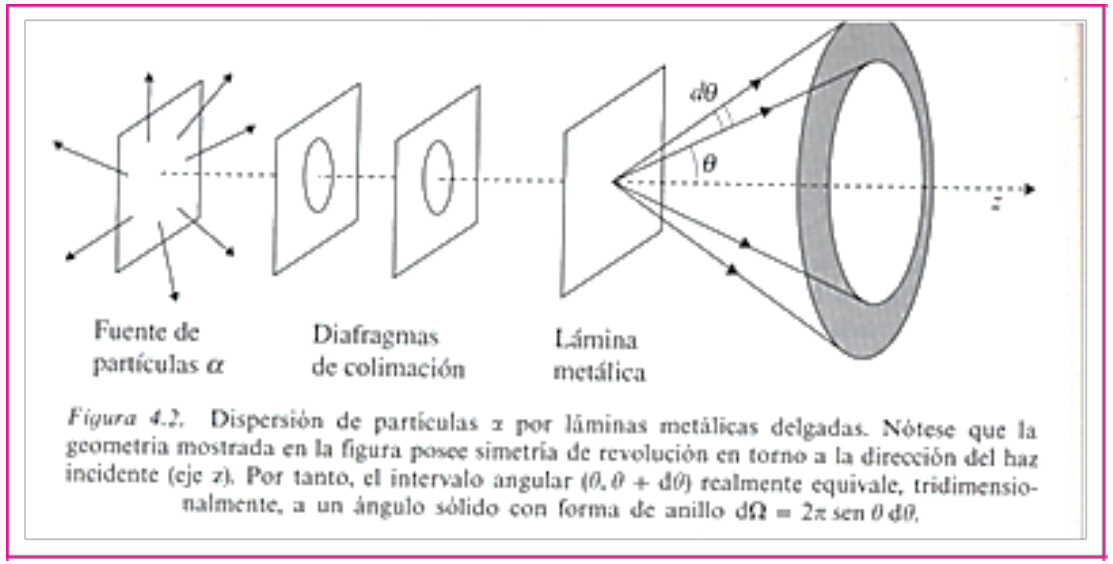


Figura 4.2. Dispersión de partículas α por láminas metálicas delgadas. Nótese que la geometría mostrada en la figura posee simetría de revolución en torno a la dirección del haz incidente (eje z). Por tanto, el intervalo angular $(\theta, \theta + d\theta)$ realmente equivale, tridimensionalmente, a un ángulo sólido con forma de anillo $d\Omega = 2\pi \operatorname{sen} \theta \, d\theta$.

Scattering of α particles

The α particle loses a small fraction of its energy when it crosses the foil.

The α particle goes across the atom and thus it provides information about the atomic structure.

In crossing the foil each α particle experiences many small deflections.

The total deflection experienced by each α particle is different.

The beam emerges from the foil as a divergent beam

Objective: To obtain a quantitative measure of its divergence.

They measured the number of α particles scattered within the angular range $\phi, \phi + d\phi$

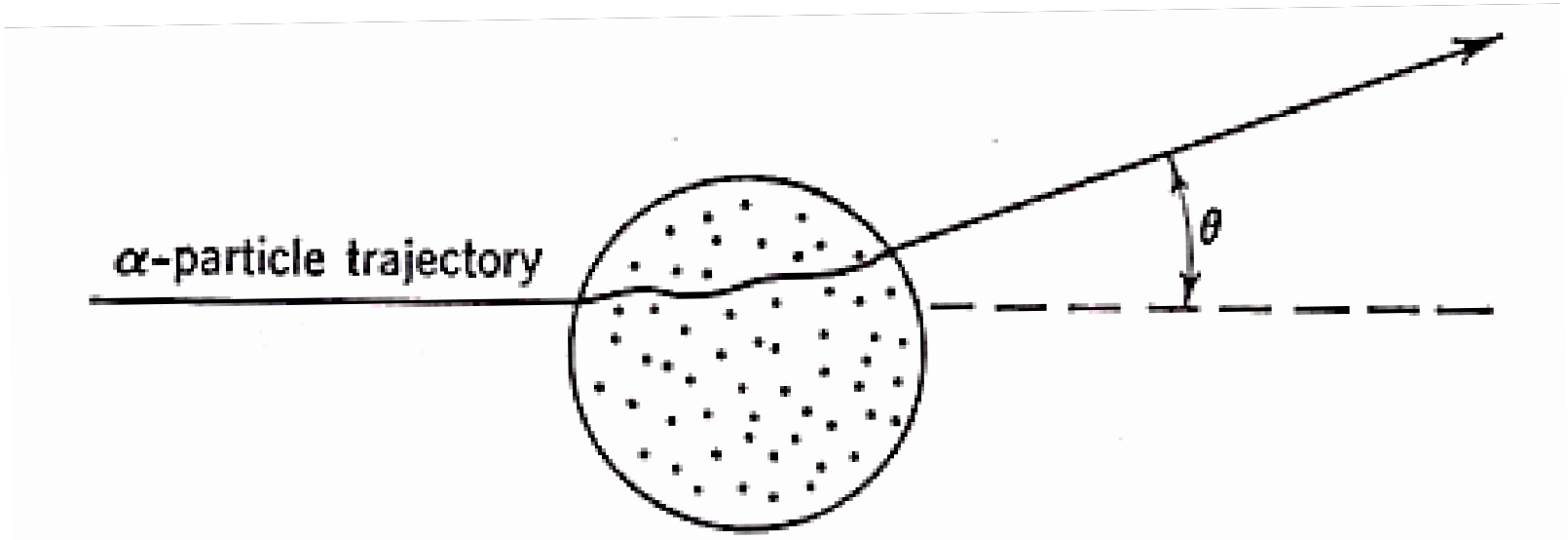
Detector: A layer of the crystalline compound ZnS and a microscope

The observer counts the total number of flashes produced per unit time as a function of the angular position of the detector.

Theoretical predictions of Thomson´s model

If we assume a distribution of electric charges in the atom, then we can calculate the scattering of an α particle.

An α particle passes through a Thomson's atom. Compute the magnitude of the scattering



An α particle passing through a Thomson model atom. The angle θ specifies the deflection of the α particle.

Theoretical predictions of Thomson´s model

Interaction with electrons

Notice $m_\alpha \approx 8000\, m_e \Rightarrow$ Scattering negligible

Head-on collision $p_e(max) \approx 2vm_e \approx \Delta p_\alpha$

$$\phi_{max} \approx \frac{\Delta p_\alpha}{p_\alpha} \approx \frac{2\mathrm{m_e v}}{\mathrm{m_\alpha v}} \approx 10^{-4}\, rads$$

Electrons as point charges ⟹ scattering event due to Coulomb interaction

Interaction with positive charge

$$F_{max} \approx \frac{Zze^2}{4\pi\epsilon_0 R^2} \qquad \Delta p_\alpha \approx F_{max}\Delta t \qquad \Delta t \approx \frac{2\mathrm{R}}{\mathrm{v}}$$

$$\phi_{max} \approx \frac{\Delta p_\alpha}{p_\alpha} \approx \frac{F_{max}\Delta t}{p_\alpha} \approx \frac{2\mathrm{zZe^2}}{4\pi\epsilon_0 \mathrm{R m_\alpha v^2}}$$

Positive charge spread throughout atom, not point charge ⟹ not a sudden kick but gentle Coulomb pull as it crosses the positive cloud.

Taking $Z \approx 50$

$$\phi_{max} \approx 10^{-4}\, rads$$

Theoretical predictions of Thomson´s model

The α particle suffers many small deflections. The total deflection angle, Φ, will be different for each particle.

Φ is a continuous random variable. Takes values within the range [0, 180º]

$$(\bar{\Phi}^2)^{1/2} = \sqrt{N}(\bar{\phi}^2)^{1/2}$$

$$\mathcal{N}(\Phi)\mathrm{d}\Phi = \frac{2I\Phi}{\bar{\Phi}^2} e^{-\Phi^2/\bar{\Phi}^2} d\Phi$$ Angular distribution of scattered α particles

$$\phi \leq 10^{-4}\, rad$$

$\mathcal{N}$ is the total number of deflections when crossing the foil

ϕ is the deflection in a single interaction.

I is the total number of incident α particles on the foil.

$\mathcal{N}(\Phi)\mathrm{d}\Phi$ is the number of α particles scattered within the angular range (Φ, Φ + *d*Φ).

Comparison with experiment

Geiger and Marsden (1909). Results with an Au foil (10^4 Å)

- Average scattering angle $(\bar{\Phi}^2)^{1/2} \approx 1° = 2\ 10^{-2}\ rad$
- More that 99% of the α : $\phi < 3°$

What about the theoretical predictions?

- $N = \frac{thickness\ foil}{atomic\ diameter} \approx \frac{10^4\ Å}{2\ Å} = 0.5\ 10^4$
- $(\bar{\phi}^2)^{1/2} = \frac{(\bar{\Phi}^2)^{\frac{1}{2}}}{\sqrt{N}} \approx 2.8\ 10^{-4}\ rad$ (Agrees with theory of 10^{-4} rad)
- Agreement with $N(\Phi)d\Phi$ when $\Phi \leq 3°$ and taking $(\bar{\Phi}^2)^{1/2} = 1°$
- But the distribution for larger angles did not agree with experiment

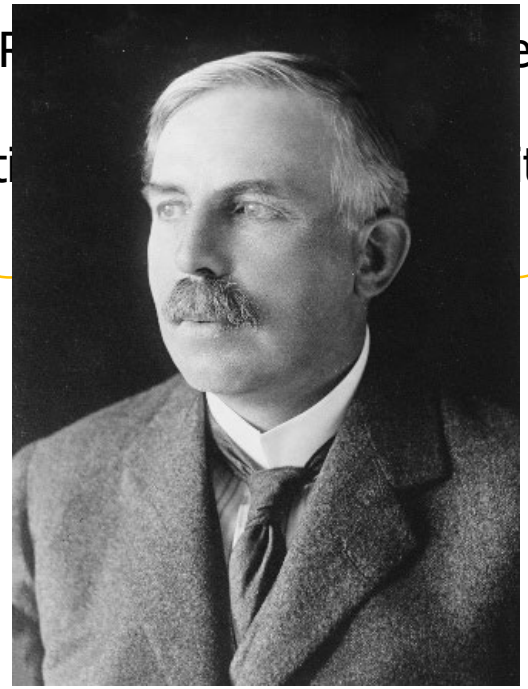
Can many tiny random deflections realistically turn an alpha particle around?

Prob($\Phi > 90°$) = 10^{-3500} (Thomson theory), 10^{-4} (Experiment)

Rutherford's atomic model

Experimental results were not consistent with Thomson's atomic model

In 1911 Rutherford proposed a new model. All the positive charge is concentrated in a small region (nucleus) surrounded by the negative electrons



Geiger and Marsden confirmed the idea in 1913

Rutherford made a detailed calculation of the angular distribution for the scattering of α particles from atoms conforming to his model.

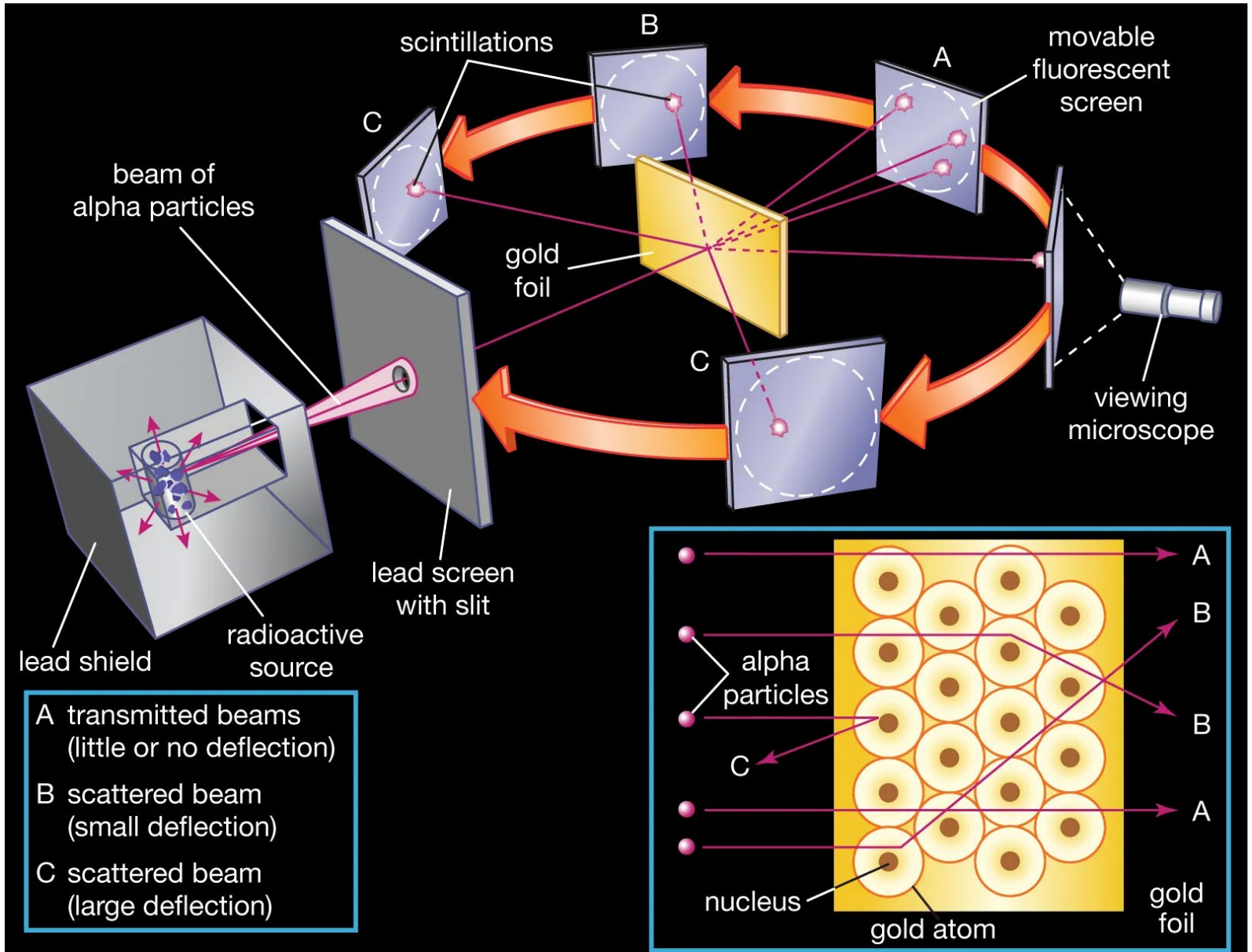
Notice that given $\phi_{max} \approx \frac{2zZe^2}{4\pi\epsilon_0 R m_\alpha v^2}$ and assuming $Z \approx 50, \mathrm{v} \approx c/20$

If $R = 1$ Å ($R = 10^{-4}$ Å) then $\phi_{max} \approx 10^{-4}\, rad$ (1 rad)

Assumptions

- Ignore the scattering due to the electrons
- The α particles do not enter the nucleus (point charges)
- The scattering is due to the repulsive Coulombic force
- The nucleus remains at rest during the scattering
- Given that $\mathrm{v}_\alpha \approx {}^{c}/_{20}$, then classical mechanics applies

Rutherford's atomic model



Rutherford's atomic model

The trajectory is on the plane that contains the nucleus. Polar coordinates (r, θ) to specify the position of the α particle

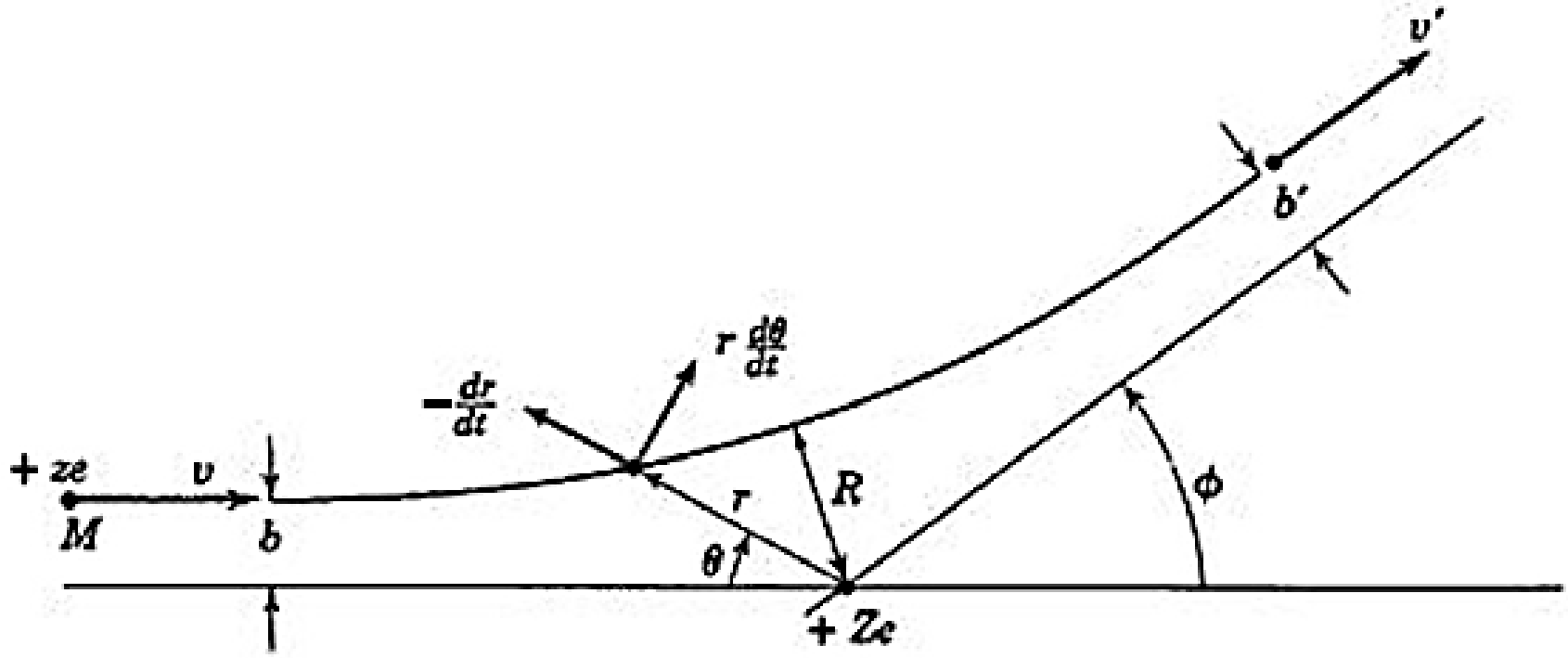


Figure 4–10. The scattering of a positively charged particle in passing near a nucleus.

Rutherford's atomic model

Polar coordinates (see Appendix D, Eisberg & Resnick)

$$\vec{v}\left(\frac{dr}{dt}, r\frac{d\theta}{dt}\right) \qquad \vec{a}\left(\frac{d^2r}{dt^2} - r\left(\frac{d\theta}{dt}\right)^2, r\frac{d^2\theta}{dt^2} + 2\frac{dr}{dt}\left(\frac{d\theta}{dt}\right)\right) \qquad \vec{F}(F(r), 0)$$

Two equations $\left(\vec{F} = m\vec{a}\right)$

$$F(r) = \frac{zZe^2}{4\pi\epsilon_0 r^2} = m\left[\frac{d^2r}{dt^2} - r\left(\frac{d\theta}{dt}\right)^2\right] \qquad 0 = r\frac{d^2\theta}{dt^2} + 2\frac{dr}{dt}\left(\frac{d\theta}{dt}\right)$$

The angular moment of the particle about the origin, *L* is constant. (Coulomb force is always in radial direction)

$$L_{initial} = L_{final} \qquad \mathrm{mvb} = \mathrm{mv'b'}$$

b: impact parameter

Energy conservation

$$\frac{1}{2}\mathrm{mv}^2 = \frac{1}{2}\mathrm{mv'}^2$$

Nucleus mass is very large so there is no recoil. Remains fixed

Rutherford's atomic model

Solve the equation

$$\frac{zZe^2}{4\pi\epsilon_0 r^2} = m\left[\frac{d^2r}{dt^2} - r\left(\frac{d\theta}{dt}\right)^2\right]$$

We are interested in the trajectory of the particle. Change variables

$$(r,\theta) \rightarrow (1/r\,,\theta) \equiv (u,\theta)$$

Search for $r = r(\theta)$

$$\frac{d^2u}{d\theta^2} + u = -D/2b^2$$

$$D = \frac{zZe^2}{4\pi\epsilon_0 (1/2)mv^2}$$

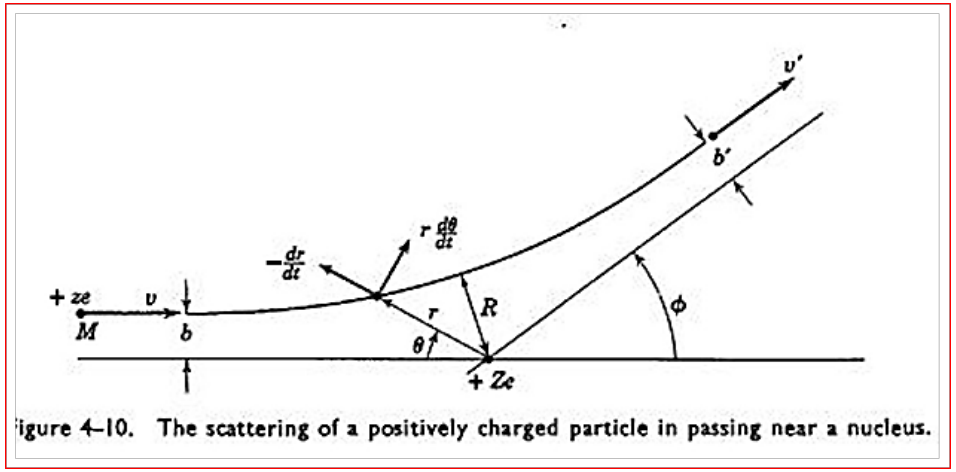


Figure 4–10. The scattering of a positively charged particle in passing near a nucleus.

D is the distance of closest approach in a head-on collision.

General solution $u = A\cos\theta + B\sin\theta - D/2b^2$

Boundary conditions When $r \rightarrow \infty$ then $\theta \rightarrow 0$ and $\frac{dr}{dt} = -\mathrm{v}$

$A = D/2b^2$ $B = 1/b$

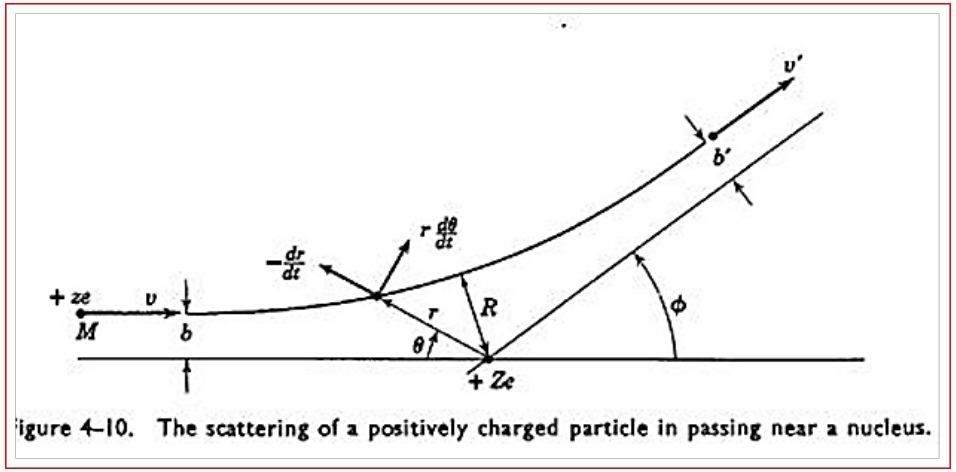
Rutherford's atomic model

Hyperbola (orbit equation) in polar coordinates

$$\frac{1}{r} = \frac{1}{b}\sin\theta + \frac{D}{2b^2}(\cos\theta - 1)$$

Scattering angle ϕ? When $r \to \infty$ $\quad \theta \to \theta'$ $\quad \theta' + \phi = \pi$

$$\frac{2b}{D} = cot\left(\frac{\phi}{2}\right)$$

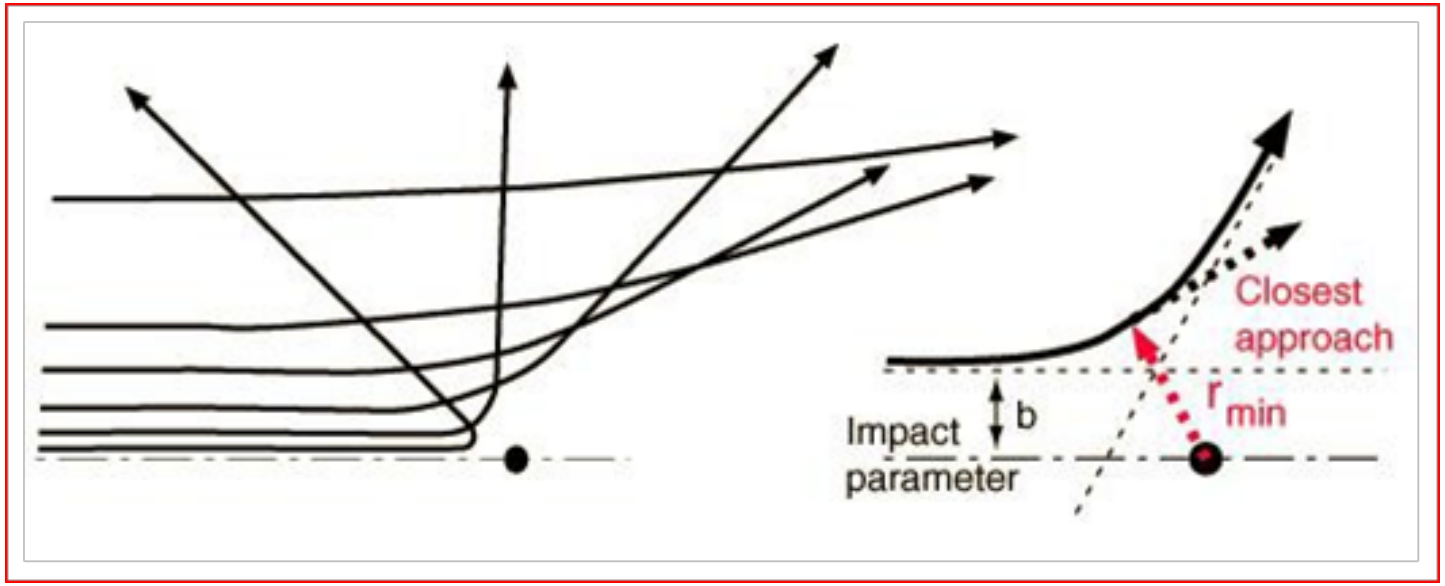


igure 4–10. The scattering of a positively charged particle in passing near a nucleus.

If the impact parameter is in the range (b, $b + db$) then the scattering angle will be within the range ($\phi, \phi + d\phi$), related by

$$db = -\frac{D}{2}\frac{1}{(\sin\phi/2)^2}\frac{d\phi}{2}$$

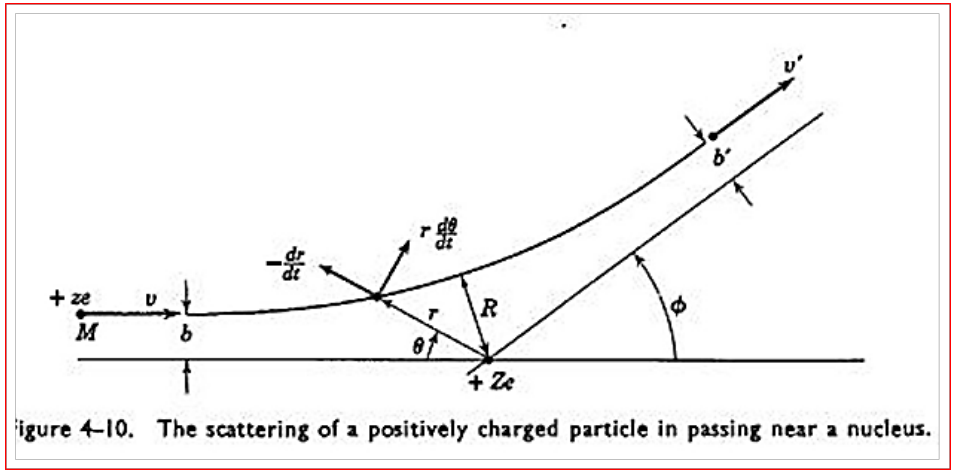
Rutherford's atomic model



Distance of closest approach *R* ?

When r = R then $\theta = \theta_R$ such that $2\theta_R + \phi = \pi$

$$R = \frac{D}{2}\left[1 + \frac{1}{\sin \phi/2}\right]$$



igure 4–10. The scattering of a positively charged particle in passing near a nucleus.

Rutherford's atomic model

An α particle with an impact parameter within the range [*b*, *b* + *db*] will be scattered within the range $[\phi, \phi + d\phi]$.

Therefore, the number of α particles scattered in $[\phi, \phi + d\phi]$, $N(\phi)d\phi$, is the same as the number of incident particles with an impact parameter within the range [*b*, *b* + *db*]

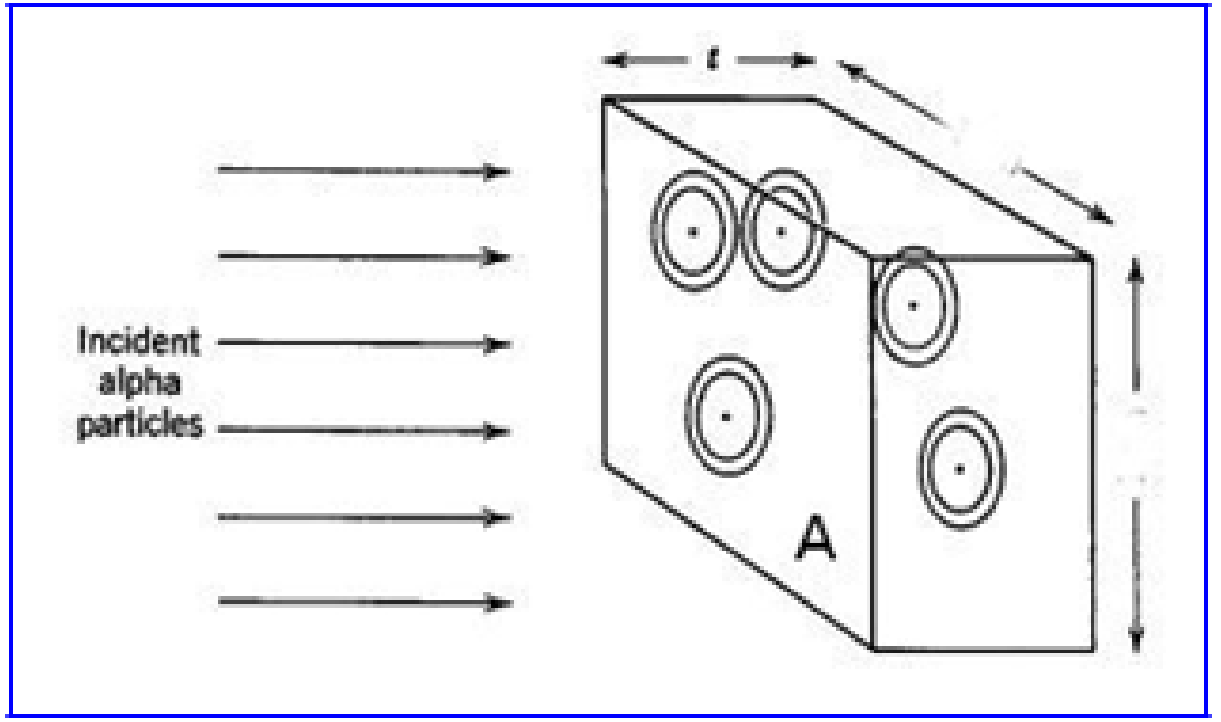
Area(ring)= $2\pi b\ db$

Prob(α particle crosses a ring)= $\frac{2\pi b db}{A}$

There are ρAt nuclei or rings

ρ is in atoms/m^3 units

Prob (α particle crosses any ring) = $\rho At \cdot 2\pi b db/A$ = $P(b)db$



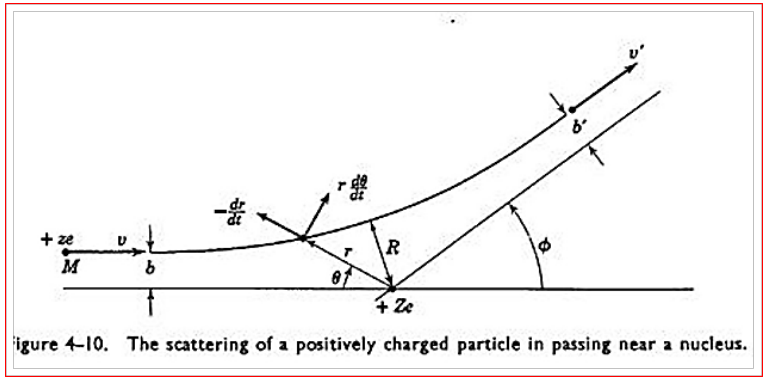
Rutherford's atomic model

P(b) db.- Probability that an α particle falls on any nucleus with an impact parameter [*b*, *b* + *db*]

$$P(b)db = -P(\phi)d\phi = -\frac{N(\phi)}{I}d\phi$$

$$P(\phi) = P(b)\left|\frac{db}{d\phi}\right| \equiv \frac{N(\phi)}{I}$$

$$N(\phi)d\phi = I\frac{\pi}{8}\rho t\left(\frac{zZe^2}{4\pi\epsilon_0(1/2)mv^2}\right)^2\frac{\sin\phi\, d\phi}{(\sin(\phi/2))^4}$$



igure 4–10. The scattering of a positively charged particle in passing near a nucleus.

I . - Number of incident α particles

$N(\phi)d\phi$.- Number of α particles scattered within $[\phi, \phi + d\phi]$

ρ.- density of the foil (in atoms/m^3)

t.- Thickness of the foil

Z e.- Total nuclear charge

Notice that the angular factor decreases with increasing angle, although it does not decrease as fast as in Thomson's model

Differential cross section

The cross section of a process measures the likelihood of that process.

A beam of α particles falling on a foil that contains *n* nuclei per unit surface, then

$$N = \sigma I n$$

n is the number of nuclei per square meter

Now, we generalize the definition. Introduce a dependence on the direction.

Specifically, the number *dN* of α particles scattered into a solid angle $d\Omega$ centered at the angle ϕ is defined as

$$dN = N(\phi)d\phi = \frac{d\sigma}{d\Omega} n I d\Omega = \sigma(\Omega) n I d\Omega$$

$\sigma(\Omega)$ is the so-called differential cross section

$\sigma(\Omega)d\Omega$ is the probability that an α particle be scattered by a nucleus, within the solid angle $d\Omega$ which is centered around ϕ

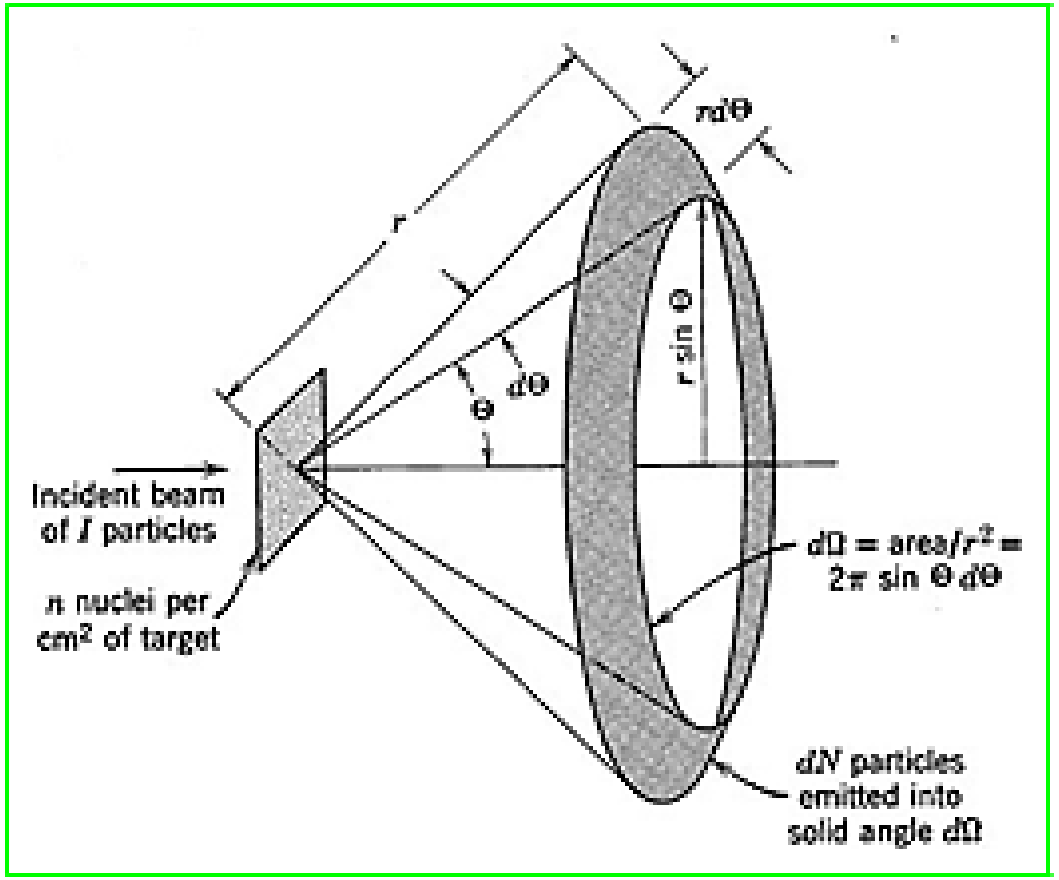
Rutherford's differential cross section

$$dN = \frac{d\sigma}{d\Omega} nI d\Omega \qquad d\Omega = \frac{2\pi r^2 \sin\phi \, d\phi}{r^2} = 2\pi \sin\phi \, d\phi$$

$$dN = N(\phi)d\phi = I\frac{\pi}{8}\rho t\left(\frac{zZe^2}{4\pi\epsilon_0(1/2)mv^2}\right)^2 \frac{\sin\phi \, d\phi}{(\sin(\phi/2))^4}$$

Identifying terms, we have Rutherford scattering differential cross section

$$\sigma(\Omega) \equiv \frac{d\sigma}{d\Omega} = \left(\frac{1}{4\pi\epsilon_0}\right)^2 \left(\frac{zZe^2}{2mv^2}\right)^2 \frac{1}{(\sin(\phi/2))^4}$$



Rutherford's formula. Comparison with experiment

$$dN = \left(\frac{1}{4\pi\epsilon_0}\right)^2 \left(\frac{zZe^2}{2mv^2}\right)^2 \frac{1}{(\sin(\phi/2))^4} Ind\Omega \qquad n = \rho t$$

$\frac{dN}{d\Omega} \propto \frac{1}{(\sin(\phi/2))^4}$ (n and energy fixed) Confirmed by using Au and Ag foils within a range [5º , 150º]

$\frac{dN}{d\Omega} \propto t$ (ϕ and energy fixed) (Remember that $n = \rho t$)

Confirmed for several elements and changing the thickness by a factor of 10

$\frac{dN}{d\Omega} \propto 1/E_k^2$ (ϕ and t fixed) Confirmed using α's of different energies within a factor of 3

$\frac{dN}{d\Omega} \propto (Ze)^2$ Used to find Z for various atoms. Equal to chemical atomic number.

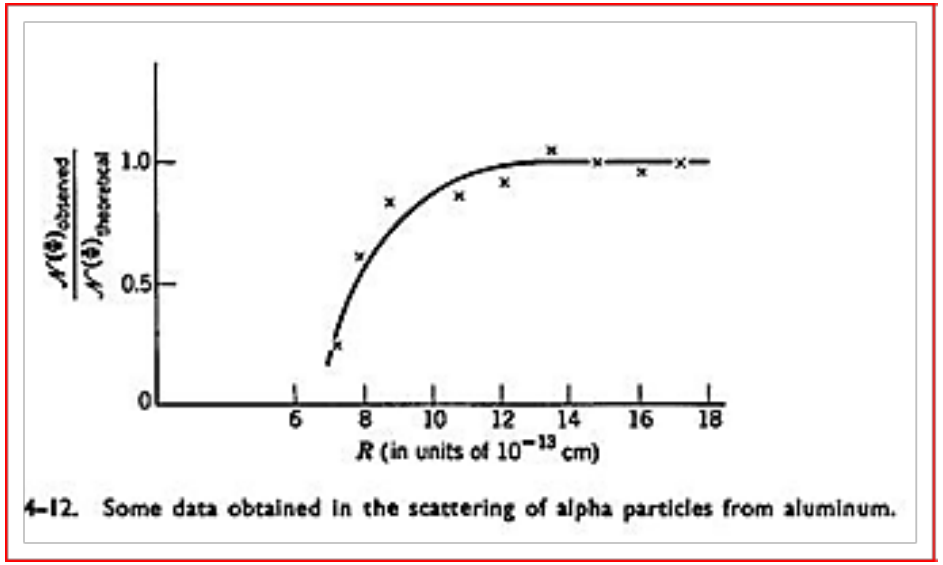
Size of the nucleus

The distance of closest approach was

$$R = \frac{D}{2}\left[1 + \frac{1}{\sin \phi/2}\right] \qquad R_{180^\circ} \equiv D = \frac{zZe^2}{4\pi\epsilon_0 (1/2)mv^2}$$

R decreases when Z decreases. How much can *R* decrease before it is smaller than the nuclear size?

In experiments with small Z atoms, they observed departures from the theoretical predictions. This was due (in part) to α particle entering the nucleus (which therefore alters the predictions).



4–12. Some data obtained in the scattering of alpha particles from aluminum.

The figure shows scattering of α particles of different energies at a fixed ϕ angle

Size of the nucleus

A more adequate probe to investigate the size of the nucleus is an electron, for two reasons:

- it is effectively point-like at the relevant length scales
- It has a charge and is not affected by the strong nuclear interaction

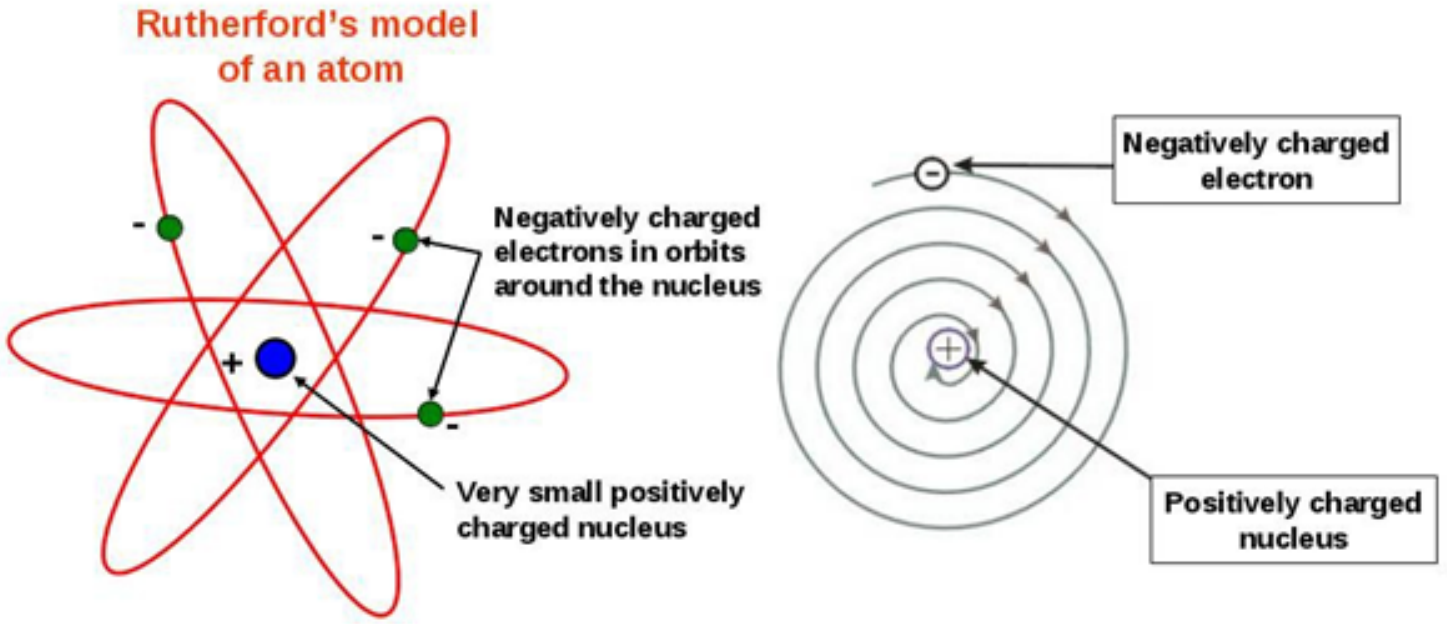
An intense research on the nuclear size was carried out in the 50s by Robert Hofstadter and coworkers: Hofstadter, R., et al., Phys. Rev. 92, 978 (1953). It was concluded that the nucleus resembled a spherical distribution of uniform positive charge and its radius can be approximated by the expression

$$r = r_0 A^{1/3} \qquad r_0 = 1.2x\,10^{-15} m = 1.2\ fm$$

Stability of Rutherford's atom

Defects of Rutherford's model of the atom :-

Any particle in a circular orbit would undergo acceleration and during acceleration the charged particle would radiate energy. So the revolving electrons would lose energy and fall into the nucleus and the atom would be unstable. We know that atoms are stable.



Stability of Rutherford's atom

No doubt about the existence of the nucleus. What about the electrons?

Are they moving around the nucleus? Forbidden by the classical electromagnetic theory.

The electron would radiate energy according to (electromagnetic theory)

$$-\frac{dE}{dt} = \frac{2}{3}\frac{e^2}{4\pi\epsilon_0 c^3}\left|\frac{d\vec{v}}{dt}\right|^2$$

For the hydrogen atom (suppose that the electron is at a distance r)

$$m_e\left|\frac{d\vec{v}}{dt}\right| = \frac{e^2}{4\pi\epsilon_0 r^2} \qquad E = -\frac{1}{2}\frac{e^2}{4\pi\epsilon_0 r} \qquad \frac{dE}{dt} = \frac{1}{2}\frac{e^2}{4\pi\epsilon_0 r^2}\frac{dr}{dt} \qquad \frac{1}{2}\frac{dr}{dt} = -\frac{2}{3}\frac{c}{r^2}r_e^2$$

$$r_e = \frac{e^2}{4\pi\epsilon_0(m_e c^2)} \approx 2.8\ 10^{-15}\ m \qquad \int_a^0 r^2 dr = -\frac{4}{3}cr_e^2\int_0^\tau dt \qquad \tau = \frac{a}{4c}(a/r_e)^2 \approx 10^{-10}\ s$$

Clasically, the atom should collapse in about 10^{-10}s .... But atoms are STABLE!

Atomic spectra. Emission

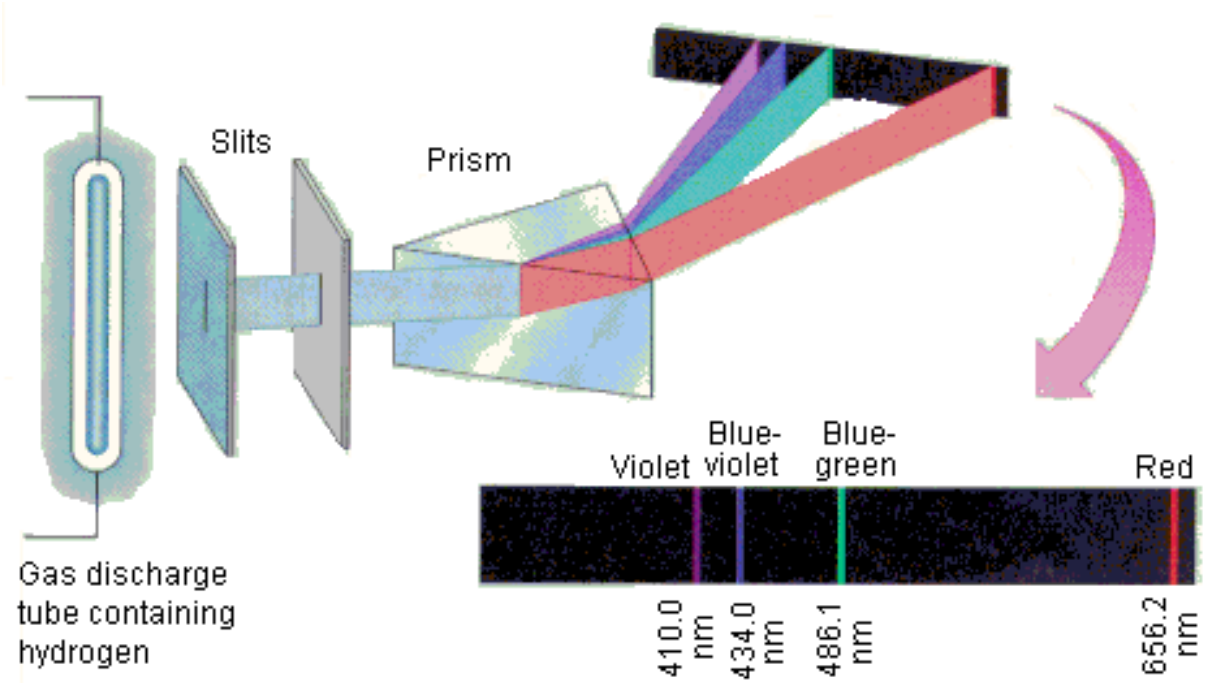
An accurate atomic model must provide an explanation for the atomic spectra, because ...

Atomic spectra inform about the internal structure of the atoms

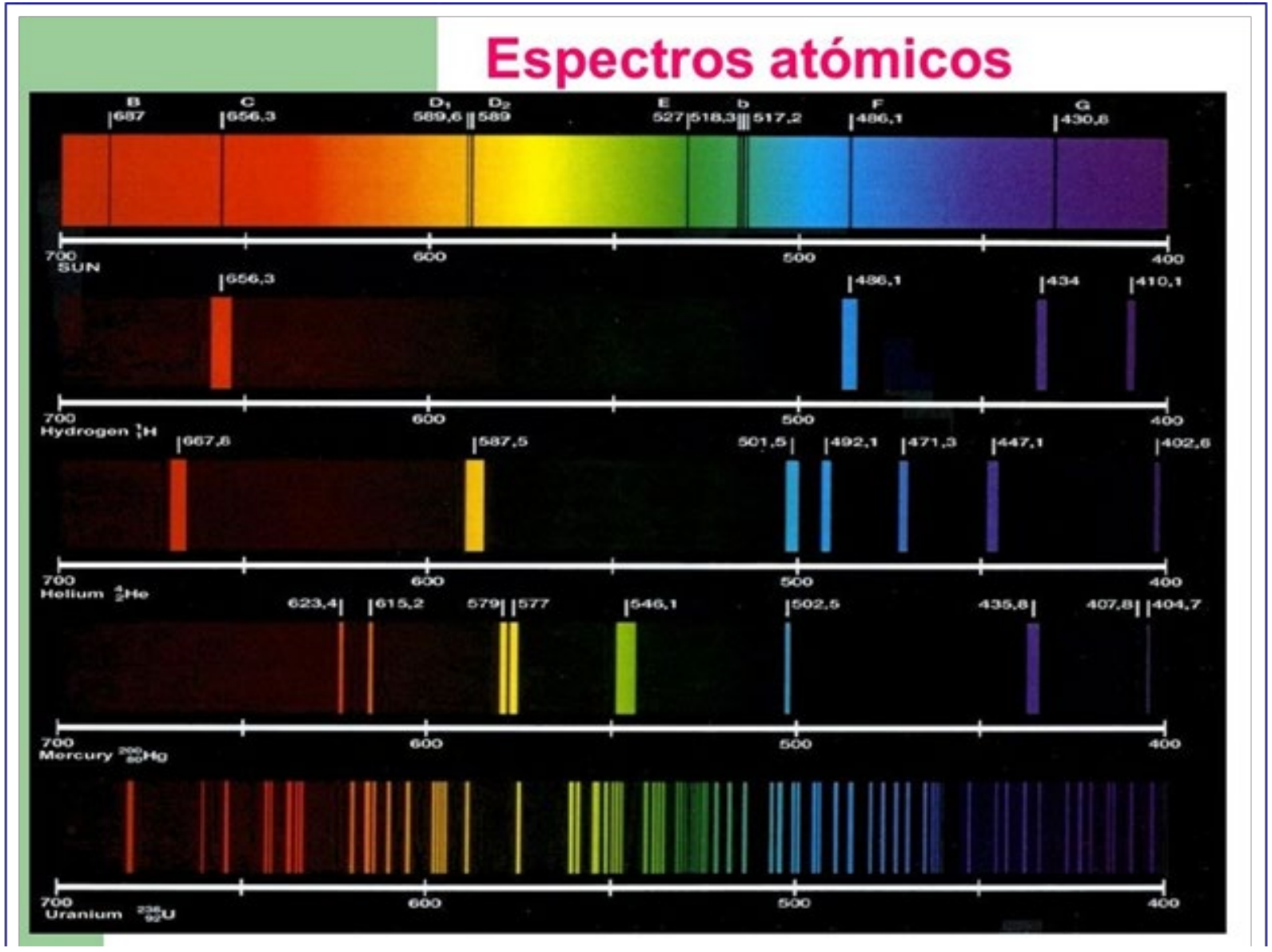
How to produce the atomic spectra?

The source is an electric discharge passing through a region containing a monoatomic gas. Then, the atoms are excited, but they return to their normal state by emitting electromagnetic radiation (emission spectrum)

The atomic spectra consist of a number of discrete wavelengths (lines) which are characteristic of each element.



Atomic spectra



Atomic spectra

Hydrogen has the simplest spectrum

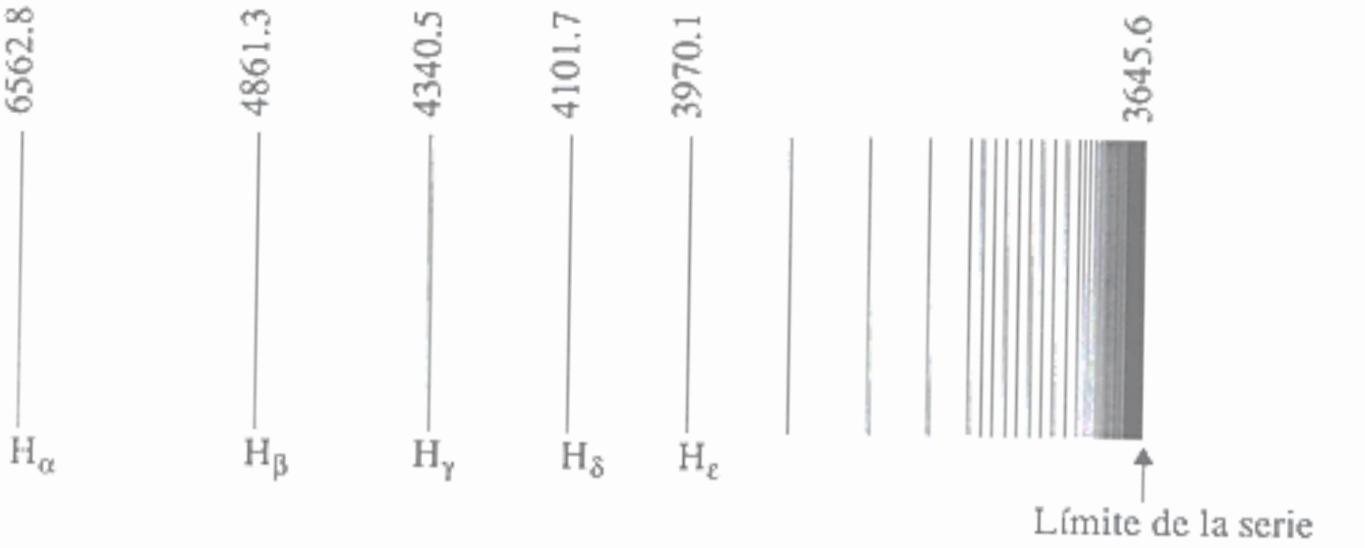


Figura 4.11. Espectro de líneas del hidrógeno atómico correspondiente a la serie de Balmer. Es corriente designar las primeras líneas mediante subíndices siguiendo el alfabeto griego

Visible part of the hydrogen spectrum. Is there a trend? Do the lines appear randomly distributed?

What seems to happen as we move toward shorter wavelengths?

Atomic spectra. Balmer series.

In 1885 Balmer found that the first four lines of hydrogen could be predicted by using the equation

$$\lambda = 3645.6\left(\frac{n^2}{n^2-4}\right)$$

where *n* is an integer number (= 3, 4, 5, 6) and λ is measured in Å.

Later on, it was found that this formula could also give the lines corresponding to the values *n* = 7, 8, 9, 10 *y* 11.

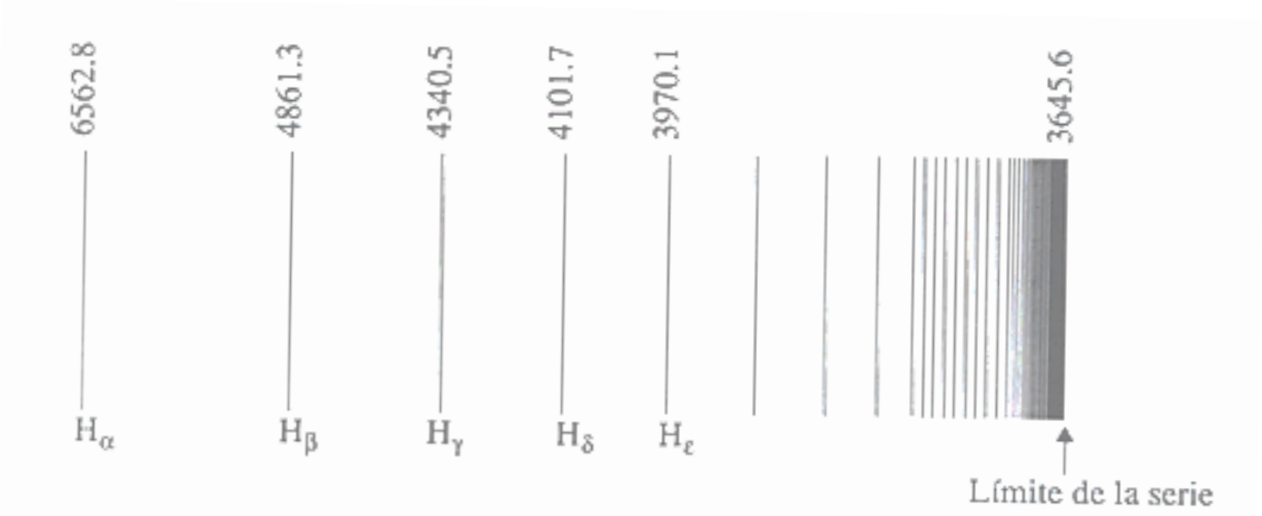


Figura 4.11. Espectro de líneas del hidrógeno atómico correspondiente a la serie de Balmer. Es corriente designar las primeras líneas mediante subíndices siguiendo el alfabeto griego

Atomic spectra

In 1890, Rydberg found convenient to deal with the reciprocal of the wavelength, $\kappa = 1/\lambda$, and the Balmer formula can be written as:

$$\frac{1}{\lambda} = \frac{1}{3645.6}\left(\frac{n^2-4}{n^2}\right) = \frac{4}{3645.6}\left(\frac{1}{4}-\frac{1}{n^2}\right) = R_H\left(\frac{1}{2^2}-\frac{1}{n^2}\right)$$

where R_H is the so-called Rydberg constant for hydrogen. Its value is

$$R_H = 10973731.539 \pm 0.0012\ m^{-1}$$

The group of all the lines obtained from the above equation is called a spectral series.

Why should integers appear in the spectrum of an atom?

Atomic spectra

All spectral series can be written as:

$$k = R_H\left(\frac{1}{m^2} - \frac{1}{n^2}\right)$$ with m fixed for each series.

Series	m	n	Region
Lyman	1	2, 3, …	UV
Balmer	2	3, 4, …	Visible
Paschen	3	4, 5, …	IR
Brackett	4	5, 6, …	IR
Pfund	5	6, 7, …	IR

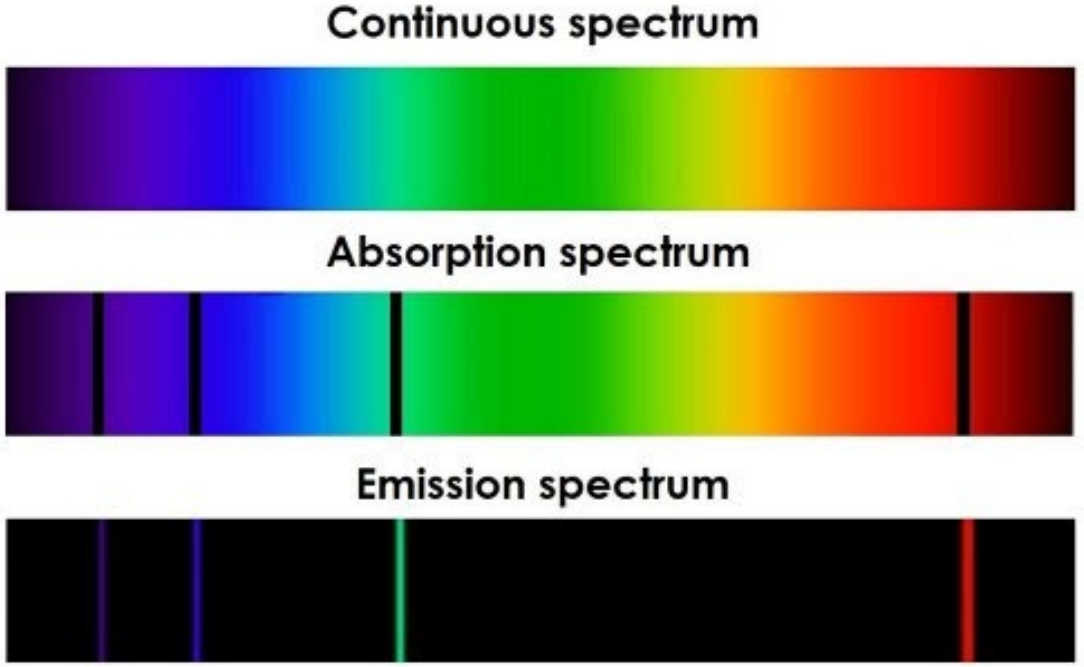
Atomic spectra. Absorption

We have seen the *emission spectrum* of an atom. A closely related property is the *absorption spectrum*

This is created by passing a continuous electromagnetic spectrum through a region containing a monoatomic gas.

Photographic plate only shows a number of unexposed lines $\Longrightarrow$ wavelengths absorbed by the atoms.

If an atom can emit a photon of wavelength λ, can it also absorb a photon of the same wavelength?



For every line in the absorption spectrum of an element, there is a corresponding (same λ) line in its emission spectrum.

The reverse is NOT true. Only certain emission lines show up in the absorption spectrum